Sets and Venn Diagrams Using Real-World Data

bfseries Instructor Lesson Plan

Math 140 – Math for Elementary Teachers

Lesson Information

Course:	Math 140 – Math for Elementary Teachers
Topic:	Applications of Sets and Venn Diagrams Using Real-World Data
Class Time:	Approximately 2 hours
Setting:	Computer lab with Microsoft Excel
Materials:	Student activity handout, simplified Excel dataset, solutions handout, computers with Microsoft Excel, and writing materials

Prior Preparation: Basic set notation, unions, intersections, complements, Venn diagrams, and basic Excel filtering.

Learning Objectives

Students will be able to:

- organize real-world data into sets and use Excel filters to identify and count observations;
- represent data using two-set and three-set Venn diagrams;
- interpret unions, intersections, and complements using data;
- solve a basic Venn diagram problem without Excel.

Lesson Plan

Time	**Activity**	Plan
10 min	**Review**	Review set notation, Venn diagrams, and the purpose of the real-world dataset.
15 min	**Part 1: By Hand**	Organize the 10-observation sample into sets and complete a two-set Venn diagram.
10 min	**Part 2: Excel**	Open the full dataset and briefly review the columns and Excel filters.
25 min	**Part 3: Two Sets**	Use Excel to find the required counts and construct the two-set Venn diagram by hand.
15 min	**Interpretation**	Use the completed diagram to interpret intersections, unions, and complements.
25 min	**Part 4: Three Sets**	Use Excel to investigate three sets and construct a three-set Venn diagram, beginning with the center intersection.
10 min	**Part 5: No Excel**	Independently complete the final Venn diagram problem without Excel.
10 min	**Reflection**	Complete the final reflection and connect sets and Venn diagrams to organizing real-world data.

Assessment

Student understanding will be assessed through **Excel filtering and data counts, completed Venn diagrams, interpretation of set notation, an independent Venn diagram application, and the final reflection.**

Assessment occurs throughout the activity as students move from organizing a small sample by hand to analyzing the larger dataset with Excel. Student work can be checked during each stage so that misunderstandings can be addressed before students move to the three-set application.

Lesson Emphasis: Excel helps students organize and examine a larger dataset, while constructing and interpreting the Venn diagrams by hand reinforces the mathematics. The goal is for technology to support students' understanding of sets rather than replace the mathematical reasoning used to construct and interpret a Venn diagram.